

ID	Citation	Source type	Window component	Circular intervention	Intended environmental benefit	Type of obsolescence	Enabling conditions	Barriers	Actors involved	Lifecycle stage	Possible unintended outcome	Classification	Suggested mitigation	Evidence type	Key finding	Page number	Study limitation
A-GS-03	Eren & Okumus, 2020	Journal article	Wooden window frame, sash, glass, frame-wall junctions	Maintenance and repair planning	Extend physical service life and avoid premature full-window replacement	Physical obsolescence	Observation-based inspection, defect classification, damage scoring, AHP-based maintenance selection	Incorrect design, incorrect construction, improper maintenance, improper use, poor communication, lack of information	Building owners, users, maintenance professionals, architects	Use and maintenance	Minor defects may become irreversible if they are not diagnosed and treated early	Implementation gap	Use regular inspections, classify defect severity, and match each defect with the appropriate maintenance action	Empirical case study	Regular maintenance and repair of wooden window systems can provide resource and energy savings when defects are diagnosed and matched with suitable maintenance actions	pp. 914-918	Applied to one historical educational building, so transferability to other materials and building types is limited
A-GS-03	Eren & Okumus, 2020	Journal article	Sealing materials, fasteners, glazing bead, hinges, drainage, frame-lintel and frame-sill junctions	Targeted component repair and replacement	Maintain window performance by repairing only damaged parts instead of replacing the whole window	Physical obsolescence	Maintenance database, defect checklist, prioritisation of repair actions	Wrong maintenance choice, poor drainage, worn seals, damaged fasteners, lack of systematic decision-making	Maintenance professionals, building owners, users	Use and maintenance	Choosing the wrong repair strategy can waste resources or fail to prevent further deterioration	Implementation gap	Use a defect-based maintenance table and prioritise interventions according to severity	Empirical case study	The study lists specific maintenance actions such as repairing seals, fasteners, glazing beads, hinges, drainage and junctions, showing that window performance can often be restored through component-level intervention	pp. 923-926	The proposed maintenance strategy was not implemented and monitored over time, so long-term performance is not verified
A-GS-06	Mohamed, 2020	Master's thesis	Insulated glazing unit, edge seal, butyl seal, spacer bar, glass panes	Remanufacturing through replaceable edge seal	Preserve high embodied-energy glass and spacer components while replacing only the weakest part	Physical obsolescence	Detachable butyl seal, hollow section, interlocking spacer, check valve for argon refill, design for on-site remanufacturing	Current IGUs are not designed for refurbishment, seal failure causes fogging and performance loss, glass often becomes landfill despite longer potential life	Designers, facade manufacturers, installers, building owners	Product design and use stage	A small weak component can determine the premature disposal of the whole glazing unit	Technological lock-in	Redesign IGUs so the edge seal can be replaced periodically without discarding the glass panes and spacer	Design thesis with prototype development	The thesis proposes a resealable IGU where the edge seal is replaced every ten years, allowing the glass panes and spacer bar to potentially remain in use for more than 100 years	PDF pp. 5, 9-10	Conceptual and prototype-based thesis, not a market-scale validated product
A-GS-06	Mohamed, 2020	Master's thesis	Float glass, coatings, foils, laminated glass, sealants, IGU cavity	Design for remanufacturing and material retention	Reduce post-consumer glass waste and avoid losing reusable glass components	Technical obsolescence	Knowledge of IGU ageing, separation of short-life and long-life components, access for remanufacturing	Adhesives, coatings, foils and sealants contaminate glass recycling streams and make current IGUs single-life products	Glass industry, facade manufacturers, designers, waste managers	End of life and remanufacturing	Circularity may fail if added performance layers make glass difficult to recycle or remanufacture	Lifecycle trade-off	Keep material layers separable and design the seal as a replaceable component rather than a permanent failure point	Design thesis and literature-based technical analysis	The thesis identifies the edge seal as the short-life component that limits IGU service life, while the glass panes and spacer have much longer potential value	PDF pp. 5, 9-10, 44-45	Focuses on IGU redesign and does not quantify full life cycle impact
A-GS-08	Femenella & Femenella, 2025	Peer-reviewed practice article	Historic steel window frames, sash, muntins, Mullions and profiles	Restoration, repair and retrofit	Extend service life while preserving original material and architectural character	Physical and desirability obsolescence	Condition assessment, preservation expertise, repair planning, shop-based restoration when needed	Assumption that historic steel windows cannot be upgraded, severe corrosion, hazardous materials, access limits, weather, budget constraints	Building owners, preservation professionals, contractors, designers	Maintenance and refurbishment	Original windows may be prematurely replaced, causing loss of character and unnecessary material replacement	Implementation gap	Assess restoration feasibility before replacement and use experienced preservation professionals	Practice-based technical article	The article argues that restoration and retrofit of existing steel windows is often feasible and can be more economical than full replacement when severe corrosion has not destroyed the members	pp. 1-3, 10	Focuses on historic steel windows, so findings are most transferable to heritage or durable metal window contexts
A-GS-08	Femenella & Femenella, 2025	Peer-reviewed practice article	Glazing pocket, hinges, sash, storm windows, weatherstripping	Non-destructive energy upgrade and retrofit	Improve thermal and water performance without replacing the whole window	Functional obsolescence	Sufficient glazing pocket depth, weight assessment, operability check, condensation control, compatible storm window design	IGUs may be too heavy, hinge stress may increase, condensation and metal incompatibility may occur	Preservation professionals, designers, contractors, building owners	Retrofit and use stage	Energy upgrades can damage original windows or reduce long-term durability if added without technical checks	Lifecycle trade-off	Check fit, weight, operability, condensation and compatibility before selecting glazing retrofit or storm windows	Practice-based technical article	Storm windows and carefully assessed glazing upgrades can improve performance, but they must be designed around the physical limits of the original window system	pp. 8-10	Does not provide quantitative LCA results
A-GS-11	Báryová et al., 2025	Journal article	Wood-based fixed and operable windows, glazing, frame, aluminium cladding, varnish	Material optimisation and maintenance-aware design	Reduce embodied environmental impact across production and maintenance stages	Physical obsolescence	LCA comparison of fixed and operable windows, coating systems, aluminium cladding and maintenance scenarios	Aluminium cladding can increase production impact, operable windows require more materials and hardware, glazing dominates many impact categories	Designers, manufacturers, LCA practitioners, building owners	Product stage A1-A3 and maintenance stage B2	Adding cladding to reduce maintenance can increase environmental burdens more than maintenance savings justify	Lifecycle trade-off	Avoid aluminium cladding where possible, prioritise fixed windows where function allows, and evaluate glazing and frame complexity carefully	Life cycle assessment	The study finds that glazing and aluminium cladding dominate the environmental profile, while operable windows have higher impacts because of additional material and hardware requirements	pp. 1, 10-14	Excludes installation, operational energy and end-of-life scenarios
A-GS-11	Báryová et al., 2025	Journal article	Wooden frame surface coating, acrylic varnish, natural oil-based varnish, paint	Protective coating and periodic maintenance	Maintain wood durability while limiting maintenance-related environmental impacts	Physical obsolescence	Scheduled coating maintenance, comparison of acrylic and natural varnish, coating quantity analysis	Natural oil-based varnish can create higher toxicity and photochemical ozone impacts, bio-based materials are not automatically lower impact	Manufacturers, designers, maintenance professionals, LCA practitioners	Maintenance stage B2	Choosing a seemingly natural or bio-based coating may shift impacts into toxicity or ozone formation categories	Burden shifting	Compare coatings across multiple impact categories and prefer water-based acrylic varnish where appropriate	Life cycle assessment	The study finds that water-based acrylic varnish has lower impacts across assessed categories, while natural oil-based varnish performs worse in terrestrial ecotoxicity and photochemical ozone creation	pp. 8, 12-14	Results depend on selected datasets, assumptions and the Slovak manufacturer-based window model
B-GS-02	Grigoryan & Daskalakis, 2025	Conference paper	High-performance facade and window system, AWS 75 SI+ components, frame, fittings, gaskets, glazing components, thermal breaks	Design for Disassembly and selective deconstruction planning	Improve material reuse, reduce construction waste, and avoid sending facade components directly to recycling or landfill	Technical obsolescence	Early design-stage disassembly planning, BIM data, bill of materials, EPD data, disassembly database, sequence IDs, recovery lists	Fixed connections, permanent sealants, non-removable thermal breaks, lack of disassembly standards, lack of supporting infrastructure	Designers, facade manufacturers, fabricators, contractors, building owners	Design, manufacturing, construction, maintenance, end of life	High-performance facade systems may be replaced entirely because future upgrades and deconstruction were not planned from the start	Implementation gap	Integrate disassembly planning into existing digital design and manufacturing workflows from the early design stage	Case study and proof-of-concept application	The study shows that disassembly planning can be integrated into facade system workflows and can generate practical outputs such as disassembly instructions, recovery lists, circularity metrics, and design improvement recommendations	pp. 706-711	Based on one manufacturer-specific window system, so broader transferability requires testing on other facade products
B-GS-02	Grigoryan & Daskalakis, 2025	Conference paper	Thermal breaks, corner cleats, permanent sealants, aluminium profiles, operable vent components	Reversible detailing and component-level recoverability	Increase the share of reusable components and reduce damage during disassembly	Technical obsolescence	Component mapping, recovery categorisation, deconstruction score, recovery score, material circularity indicator, disassembly time calculation	Some components cannot be removed without damaging adjacent elements, especially rolled thermal breaks, permanent sealants and petroleum-derived materials	Product designers, facade system manufacturers, fabricators, contractors	End of life and future design	A system may appear demountable but still contain hidden locked-in components that prevent high-value reuse	Technological lock-in	Redesign critical connections and short-life components so they can be accessed, removed and replaced without destroying surrounding parts	Case study with disassembly metrics	The AWS 75 SI+ case identifies reusable, recyclable and waste components, but also shows that some critical components remain unrecoverable because of connection and material choices	pp. 711-712	Results depend on assumptions about one skilled worker, one workstation and selected system configurations
B-GS-05	Aktas & Charubab, 2024	Conference paper	Aluminium-framed window, glazing, frame parts, fasteners and component interfaces	Robotic disassembly for window upgrading	Support repair, replacement, recycling and remanufacturing while extending window service life	Functional and technical obsolescence	Disassembly Map, 3D model, section drawings, robotic path planning, virtual simulation, UR/life robotic demonstration interface	Most windows are not designed for end-of-life disassembly, and detailed geometric documentation is often missing before dismantling	Designers, architects, engineers, product designers, researchers, robotic automation specialists, contractors	End of life and upgrading	Without clear disassembly sequences, useful components may be damaged or sent to lower-value recovery routes	Implementation gap	Create disassembly maps from detailed drawings or 3D models before dismantling, then use them to guide manual or robotic disassembly	Methodology and robotic demonstration	The study shows that disassembly maps can define the sequence of actions needed to dismantle window components and can support systematic manual or robotic upgrading	pp. 423-428	Demonstrated on a scaled aluminium window model rather than a full-size window in a real building
B-GS-05	Aktas & Charubab, 2024	Conference paper	Window components with reusable high residual value	Selective disassembly	Save disassembly time, increase component recovery and support repair or maintenance rather than full disposal	Physical and functional obsolescence	Identification of reusable parts, ordered disassembly sequence, part removal direction, target automation level	Manual labour intensity, lack of standardized methods, variable component quality, missing documentation	Designers, architects, engineers, contractors, robotic system developers	Maintenance, repair and end of life	If disassembly is too slow or uncertain, repair and reuse may become economically unattractive	Implementation gap	Use selective disassembly for the specific components needed for repair, upgrade or reuse instead of fully dismantling the whole system	Methodological conference paper	Selective disassembly is presented as useful for repair, maintenance and end-of-life recovery because it targets specific components while reducing time and effort	pp. 424-426	The study focuses on method validation and does not quantify full environmental savings
B-GS-10	Rose, 2019	Doctoral thesis	Existing building components, reclaimed construction components, timber components, potential reusable elements	Component management, triage, reuse, repurposing and upcycling	Retain embodied impacts and avoid value loss through downcycling or disposal	Functional and economic obsolescence	Early identification of waste streams, component audit, information system, local reuse network, creative redesign, storage and intermediary actors	Lack of quantity and quality certainty, weak reclamation infrastructure, missing remanufacturing and recertification enterprises, conventional waste logistics	Clients, contractors, designers, waste management companies, reuse enterprises, local authorities	End of life and pre-demolition phase	Components with remaining value may enter waste management because no system exists to identify and route them to higher-value reuse	Implementation gap	Use an urban triage process before demolition to separate components suitable for reuse, repurposing or upcycling from those better suited to downcycling	Doctoral thesis with mixed methods and case studies	The thesis argues that construction needs component management rather than only waste management, because direct reuse, repurposing and upcycling can preserve more value than normal recycling routes	pp. 5, 89-102, 234	Broader construction-component focus, so transfer to residential window systems is conceptual rather than direct
B-GS-10	Rose, 2019	Doctoral thesis	Existing building stock as a material bank, removable building components, secondary timber example	Existing Buildings as Material Banks and information-based triage	Improve future reuse by knowing what components exist, where they are, and what condition they are in	Functional obsolescence	Material audits, stock information, pre-development data capture, reuse marketplaces, storage options, third-party intermediaries	Missing information about existing buildings, late demolition-stage decisions, damaged materials during waste handling, lack of demand matching	Building owners, designers, demolition contractors, reuse platforms, municipalities	Pre-demolition, end of life and future reuse	Materials may technically be reusable but become unusable because they are identified too late or damaged during removal and handling	Implementation gap	Capture component information earlier and connect supply projects with demand projects before demolition or strip-out begins	Doctoral thesis with urban-level case studies	The thesis frames information and timing as core barriers to component reuse, especially where materials only become visible once they are already entering waste streams	pp. 106-120	Not window-specific and therefore used mainly for the system-level reuse logic
B-GS-14	Yang et al., 2024	Journal article	Modular building component, steel frame core, concrete slab, steel connector	Multi-cycle reuse with value retention processes	Maximise environmental benefits across multiple use cycles rather than only the first use	Technical obsolescence	Design for Deconstruction, direct reusability, repair, replacement, recyclability, sensitivity analysis, LCA allocation rules	Future reuse rates, repair needs, replacement of subcomponents, uncertain recyclability, uncertain transport and end-of-use scenarios	Designers, contractors, LCA practitioners, building owners	Production, reuse cycles, repair, replacement, recycling and disposal	Reuse may still create environmental loads when repair or replacement activities are needed to enable the next cycle	Lifecycle trade-off	Prioritise direct reuse and repair over replacement, and design core components so they remain reusable with minimal value retention activity	Life cycle assessment with sensitivity analysis	The study finds that direct reusability and recyclability are important for environmental benefit, while replacement of subcomponents should be avoided because it increases impacts	pp. 543-545, 557	Case study is modular building components rather than windows or façades
B-GS-14	Yang et al., 2024	Journal article	Core components and replaceable subcomponents	Design for direct reuse and reduced replacement dependency	Avoid additional virgin material demand and disposal burdens in later cycles	Technical obsolescence	Distinction between non-replaceable core components and replaceable subcomponents, probabilistic LCA, value retention process modelling	Repair processes can affect core-component impacts, while subcomponent replacement is especially impact-sensitive	Structural designers, engineers, architects, manufacturers, LCA practitioners	Intermediate reuse cycle and end-of-use treatment	A circular design can lose part of its benefit if reuse depends on repeated replacement of subcomponents	Burden shifting	Design subcomponents for repairability and durability, and minimize replacement-intensive reuse scenarios	LCA and global sensitivity analysis	The study shows that the environmental impact of reusable components depends not only on reuse itself but also on the repair and replacement activities required to make reuse possible	pp. 545, 556-557	Economic and social impacts are not included in the main assessment
B-GS-19	Järvelä et al., 2025	Journal article	Building components, material layers, assemblies, structural joints	Disassembly potential assessment	Support reuse by making disassembly measurable during early design	Technical obsolescence	Streamlined DP method, material quantity data, material-layer assessment, expert validation, case studies	Existing DP methods often need detailed information that is unavailable in early design, and many circularity methods are not standardized	Designers, LCA practitioners, architects, engineers, demolition experts	Early design and end-of-life planning	Reuse potential may be ignored because disassembly cannot be assessed early enough to influence design decisions	Implementation gap	Use a simplified material-layer-dependent DP assessment that can work with data available during early design	Journal article with method development and case studies	The article proposes a practical disassembly potential method that can be used earlier in design and aligned more easily with LCA workflows	pp. 634-637	Developed in a Finnish context and focused on buildings and components generally rather than windows specifically

B-GS-19	Jarvela et al., 2025	Journal article	Connections, assemblies, material layers and building components	Early-stage Design for Disassembly evaluation	Reduce future demolition waste and extend material lifecycles	Functional obsolescence	Quantitative assessment, simplified parameters, BIM-compatible data, expert comparison	Designers still need manual work for DP assessment, BIM proposals remain partly theoretical, and available data varies by design phase	Designers, BIM specialists, consultants, practitioners, construction industry actors	Early design, detailed design and future end of life	If DP assessment remains too complex, circular design may remain theoretical rather than affecting real design decisions	Implementation gap	Keep DP assessment simple enough for actual design workflows and connect it with the same data used for LCA	Method-development journal article	The study argues that DfD must be considered from early design phases to maximise greenhouse gas reductions from future reuse	pp. 634-637	It does not test the method specifically on residential window replacement or repair scenarios
B-GS-20	Droste, 2023	Master's thesis	Aluminium curtain wall façade system, glass, aluminium profiles, EPDM gaskets, thermal breaks, silicone sealants	Disassembly assessment framework for façade reclamation	Increase material recovery, reduce disassembly time and support high-value end-of-life routes	Technical obsolescence	Disassembly potential, reclamation potential, disassembly time, material database, process database, connection database, case study assessment	Key information is often unstructured, subjective or unavailable to computational tools and façade systems are highly multi-material and connection-dependent	Designers, façade engineers, manufacturers, consultants, demolition experts, reclamation auditors	Design and end of life	A façade may have theoretical recovery potential but remain difficult to reclaim because the required data is not structured or accessible	Implementation gap	Build structured databases for materials, processes and connections, and use material passports to assess disassembly and reclamation potential	Master's thesis with framework demonstration	The thesis develops a façade-specific framework that evaluates how design choices influence material reclamation potential at the end of life	pp. 5, 13-16, 95-110	Tested on an aluminium curtain wall case study, so further validation is needed for other façade typologies
B-GS-20	Droste, 2023	Master's thesis	Façade connections, adhesive sealants, laminated glazing, aluminium profiles, gaskets, thermal breaks	Standardised connection design and material passport support	Reduce barriers to repair, reuse, remanufacturing and recycling	Technical obsolescence	Material passports, BIM, asset information models, digital information models, connection data, lifecycle information, standardized content, reverse logistics	Lack of technical composition data, no certification tools for reusable elements, missing protocols, weak reverse logistics, adhesive layers, high customization	Manufacturers, designers, contractors, logistics managers, reclamation auditors, recycling companies	Design, maintenance, reverse logistics and end of life	Highly customized or adhesive-based façade products can become locked into destructive recycling or disposal despite material value	Technological lock-in	Standardize connection types, reduce unnecessary customization, record component data in material passports, and avoid inseparable adhesive layers where possible	Master's thesis and case study framework	The thesis identifies data, certification, protocol and logistics barriers as major reasons why building elements are not reclaimed at high value even when disassembly potential exists	pp. 23-26, 38-43, 116-117	Some recommendations are conceptual and require industry validation
C-GS-01	Kaddoura et al., 2019	Journal article	Not window-specific, transferable to residential window systems as passive durable products	Product lifetime extension through repair, refurbishment and service-based models	Reduce production-related impacts by delivering more use from the same product or component	Physical obsolescence	Product-service system, repair and refurbishment capacity, economically viable service offer, passive product with production-dominated impacts	Company willingness, administration costs, labour costs, reverse logistics, need for business viability	Manufacturers, service providers, users, companies, repair and refurbishment actors	Use, maintenance and refurbishment	Repair or refurbishment may fail if the service model is not financially viable or if operational service burdens become too high	Implementation gap	Focus lifetime-extension strategies on passive durable components where production impacts dominate and where repair and refurbishment remain practical	Multiple case study with LCA and LCC	The study finds that extending the lifetime of passive durable products through repair or refurbishment reduced life-cycle GWP in all five cases, mainly because extraction and production dominated the impacts	pp. 1-3, 17-18	Not construction- or window-specific, so transfer to residential windows must be made carefully
C-GS-01	Kaddoura et al., 2019	Journal article	Not window-specific, transferable to window frames, glazing units and hardware as passive durable components	Repeated repair or refurbishment	Increase number of uses and reduce impact per unit of function	Physical obsolescence	Sensitivity analysis, knowledge of the highest-impact components, repair and refurbishment limits	Risk of repairing too often, rebound effects, uncertainty in large-scale adoption, toxicity data gaps for some materials	Manufacturers, service providers, users, LCA practitioners	Use, maintenance and refurbishment	Assuming repair is always environmentally positive may lead to excessive repair cycles or underestimated burdens	Rebound effect	Identify the environmental hotspots and avoid exceeding the useful repair and refurbishment limit	LCA and LCC case study	The authors note that lifetime extension improves results up to a limit and that companies should avoid assuming unlimited repair or refurbishment is automatically beneficial	pp. 17-19	Rebound effects and large-scale adoption are discussed as future research rather than fully quantified
C-GS-02	Zink & Geyer, 2017	Journal article	Not component-specific, transferable to reused or refurbished window systems and components	Reuse, refurbishment, remanufacturing and recycling	Substitute secondary production for primary production and reduce environmental impacts	Functional obsolescence	Actual displacement of primary products, secondary products that are good substitutes, market demand for reused components	Poor substitutability, lower perceived quality, weak market displacement, niche markets that add consumption instead of replacing new production	Producers, consumers, refurbishers, recyclers, market actors, policymakers	End of life and secondary use	Reused or refurbished products may be produced in addition to new products rather than replacing them	Rebound effect	Evaluate whether reused or refurbished window components actually displace new window production before claiming environmental benefit	Conceptual and theoretical article	The article defines circular economy rebound as the situation where lower-impact secondary production increases total production or consumption, reducing or cancelling the intended environmental benefit	pp. 1-5	Conceptual paper, does not provide a building-specific empirical case
C-GS-02	Zink & Geyer, 2017	Journal article	Not component-specific, transferable to secondary window-product markets	Secondary production through repair, reuse or recycling	Reduce primary material extraction and product manufacturing	Functional and economic obsolescence	Strong substitution between secondary and primary goods, low buyer price sensitivity, careful market positioning	Price effects, lower prices for secondary goods, increased consumption, partial displacement, niche market expansion	Producers, consumers, secondary-product producers, policymakers	Secondary market and end-of-life stage	Cheaper circular products may increase total demand or shift consumption towards reducing primary production	Rebound effect	Focus on displacement potential rather than simply closing loops, and avoid circular strategies that rely mainly on low prices to attract demand	Conceptual and theoretical article	The authors argue that simply closing material loops is not enough because environmental benefit depends on whether secondary products actually displace primary production	pp. 5-8	The paper gives general mechanisms, so project-specific market evidence would still be needed for residential windows
C-GS-04	Castro et al., 2022	Journal review article	Not component-specific, transferable to circular window repair, spare-part and reverse-logistics systems	Circular economy implementation and transition	Reduce waste and emissions through repair, reuse, remanufacturing and recycling	Functional obsolescence	Systems thinking, early detection of rebound risks, infrastructure planning, governance, stakeholder coordination	Additional spare parts, new logistics infrastructure, transport, disassembly processes, new sales points, operating costs	Policy-makers, companies, consumers, service providers, logistics actors	System implementation, maintenance and end of life	A circular repair or reuse system may create new material flows, transport needs and infrastructure burdens that reduce the intended benefit	Rebound effect	Assess the transition system early, including spare parts, transport, logistics and additional infrastructure before claiming circular benefit	Systematic literature review	The review shows that CE strategies can trigger rebound through the transition process itself, because circular systems may require extra infrastructure, transport, spare parts and disassembly activity	pp. 1-6	Broad CE review, not construction- or window-specific
C-GS-04	Castro et al., 2022	Journal review article	Not component-specific, transferable to circular service models for window repair and replacement	Product-service systems, reuse markets and circular business models	Keep products and materials in use while reducing primary production	Functional and desirability obsolescence	Behaviour-aware design, policy support, sustainable research and development, price control, early rebound detection	Consumer behaviour, increased access, lower price, new markets, additional consumption, business models that do not change linear habits	Consumers, companies, policymakers, service providers	Use, market adoption and secondary production	Circular services may increase access and consumption instead of reducing material demand	Rebound effect	Combine circular design with behavioural measures, early detection tools, policy support and market controls where needed	Systematic literature review	The review identifies initiating, developer and mitigating mechanisms of circular rebound, including CE strategies, CE transition, innovation, business models and environmental policy	pp. 4-9	The paper synthesizes literature rather than testing a specific product system
C-GS-10	Corona et al., 2019	Journal review article	Not component-specific, transferable to circularity assessment of residential window repair, reuse and recycling scenarios	Circularity measurement and assessment	Support evidence-based circular decisions and avoid false environmental claims	Functional obsolescence	Comprehensive metrics, LCA, multiple indicators, clear CE definition, assessment of environmental, economic and social effects	Narrow material-flow indicators, inconsistent metrics, lack of social and economic assessment, poor treatment of multiple cycles and downcycling	Designers, researchers, LCA practitioners, policymakers, companies	Design assessment and end-of-life evaluation	A window strategy may look circular because material loops are closed while still shifting burdens to other environmental or social categories	Burden shifting	Use circularity assessment together with LCA or broader sustainability indicators instead of relying only on material circularity scores	Literature review and critical assessment	The review finds that many circularity metrics do not address the full CE concept and may lead to burden shifting from reduced material consumption to other environmental, economic or social impacts	pp. 1-4, 9-10	Review focuses on metrics broadly, not window-specific assessment
C-GS-10	Corona et al., 2019	Journal review article	Not component-specific, transferable to end-of-life window recycling and reuse assessment	Circularity metrics, LCA allocation and multi-cycle assessment	Correctly account for benefits and burdens of recycling, reuse and secondary material use	Functional obsolescence	Clear allocation rules, attention to downcycling, multiple-use cycles and primary and secondary product relationships	Inconsistent allocation of recycling credits, underrepresentation of multiple cycles, unclear treatment of downcycling, oversimplified circularity scores	LCA practitioners, policymakers, designers, manufacturers, recyclers	End of life, secondary production and future cycles	Environmental benefits may be overstated if recycling credits are allocated unclearly or if downcycling is ignored	Burden shifting	State allocation assumptions clearly and account for multiple cycles and material-quality loss when assessing window recycling or reuse	Literature review and critical assessment	The review argues that circularity metrics should clearly indicate how recycling benefits are allocated between primary and secondary products and should account for downcycling and multiple cycles	pp. 1, 6-10	The paper evaluates measurement tools rather than providing direct empirical evidence for windows
D-GS-01	Carlisle & Friedlander, 2016	Journal article	Aluminium, wood, aluminium-clad window and PVCu window frames, coatings, weatherstripping, sealants and hardware	Service-life extension through durability and maintenance modelling	Reduce full-frame replacement and lower life-cycle impacts over the building lifespan	Physical and functional obsolescence	Realistic service-life assumptions, maintenance scenarios, material-specific durability data, full-life LCA modelling	Lack of reliable service-life data, cradle-to-gate EPDs excluding use phase, uncertainty around maintenance regimes and replacement timing	Architects, LCA practitioners, manufacturers, building owners, designers	Product stage, use stage and end of life	Incorrect assumptions about service life can strongly distort the perceived environmental performance of a window frame material	Lifecycle trade-off	Model multiple service-life and maintenance scenarios instead of relying on warranty periods or cradle-to-gate data only	Comparative life cycle assessment	The study finds that expected service life is one of the most important factors in assessing window frame environmental impact, because one full replacement can effectively double product-stage impacts	pp. 1645-1651	Glazing is excluded and service-life assumptions remain uncertain
D-GS-01	Carlisle & Friedlander, 2016	Journal article	Aluminium and other recyclable window frame materials	End-of-life recycling and material recovery modelling	Promote recovery of high-value materials and reduce future virgin material demand	Technical obsolescence	End-of-life recycling method, Module D logic, recovery rates, recycling credit allocation	Uncertain future reclamation rates, methodological disagreement, difficulty comparing multi-material assemblies	LCA practitioners, designers, manufacturers, recyclers, policy actors	End of life and future material cycles	Environmental benefits may be overstated or understated depending on how recycling credits are allocated	Burden shifting	State recycling allocation method clearly and use sensitivity analysis for recovery rates and end-of-life assumptions	Comparative life cycle assessment	The study shows that recycling rates strongly influence aluminium frame results and that end-of-life modelling choices can change environmental recommendations	pp. 1645-1653	Future recovery rates are predicted rather than observed
D-GS-02	Sällström Eriksson & Lidelow, 2025	Journal article	Existing wooden windows in a 1980s multi-residential building, replacement wooden, aluminium-clad and PVC windows	Maintenance versus replacement	Minimise life-cycle climate impact and costs by prolonging existing window service life	Physical obsolescence	Periodic maintenance, LCA-GWP, LCC, 60-year reference study period, sensitivity analysis	Owner preference for replacement, perceived certainty of energy savings, high investment costs, uncertain heating emission factors and service lives	Property owners, housing companies, designers, contractors, policymakers, LCA practitioners	Maintenance, replacement and operational heating stage	Replacing windows may reduce operational heating energy but still increase life-cycle impacts or costs because of embodied impacts and investment costs	Lifecycle trade-off	Compare maintenance and replacement using both embodied and operational impacts before choosing full replacement	LCA and LCC case study	The study finds that maintaining existing windows was generally preferable for both life-cycle climate impact and costs, with replacement beneficial only under specific conditions	pp. 766-786	Focuses on one Swedish multi-residential building in a subarctic climate
D-GS-02	Sällström Eriksson & Lidelow, 2025	Journal article	Window intervention options affecting heating demand and embodied impacts	Sensitivity-based decision support for maintenance and replacement	Avoid making replacement decisions based only on assumed operational energy savings	Functional obsolescence	Sensitivity analysis of district heating emission factor, heating price, U-value, reference study period, service life and discount rate	High uncertainty in local heating context, future energy prices, initial window thermal performance and intervention frequency	Housing companies, property owners, energy consultants, LCA practitioners	Operational energy, product stage and decision-making stage	Energy-efficient replacement can look beneficial under one heating scenario but not under another	Burden shifting	Test uncertain parameters before claiming that replacement is the more sustainable option	LCA and LCC sensitivity analysis	The study finds that LCA-GWP results are more sensitive than LCC results and that replacement is mainly justified under climate-intensive and expensive heating conditions	pp. 776-784	Results are limited to climate impact and costs, not broader environmental categories
D-GS-03	Sällström Eriksson & Lidelow, 2025	Journal article	Windows in two residential buildings from 1910 and 1985 in subarctic Sweden	Context-specific maintenance, replacement and secondary glazing assessment	Balance embodied impacts, operational heating savings and renovation costs	Physical and functional obsolescence	Building-specific LCA-GWP, LCC, energy simulation, sensitivity analysis, comparison of building characteristics and local context	No single best intervention, different initial thermal performance, different heating emission factors, different climate zones and transportation conditions	Property owners, housing companies, designers, energy consultants, LCA practitioners	Product stage, installation, operational heating and maintenance	A replacement strategy that works for one building may increase impacts or costs in another building	Implementation gap	Adapt window intervention decisions to each building instead of applying a universal replacement rule	Comparative LCA and LCC study	The study finds that building characteristics and local context strongly affect whether maintenance or replacement is preferable, so replacing windows does not guarantee lower life-cycle climate impact or costs	pp. 1-6, 20-21	The two buildings are illustrative and not directly generalizable

D-GS-03	Sällström Eriksson & Lidclow, 2025	Journal article	Existing windows, replacement windows and secondary glazing options	Maintenance and low-material intervention strategy	Reduce material use while maintaining acceptable energy performance	Physical obsolescence	Initial thermal-property assessment, district heating emission factor, climate-zone sensitivity, intervention-rate testing	Low-carbon heating makes operational savings less valuable, while high embodied impacts and repeated interventions can dominate results	Property owners, contractors, designers, LCA practitioners	Maintenance, replacement, secondary glazing and operational stage	Operational energy savings may not compensate for embodied impacts, especially in buildings with good initial performance or low-carbon heating	Lifecycle trade- off	Prefer maintenance or low- material interventions when existing performance is acceptable and heating emissions are low	Comparative LCA and LCC study	The study concludes that non-energy-related renovation such as maintaining original windows can be the most climate-friendly option in buildings with good initial thermal performance and low-carbon district heating	pp. 11-12, 20- 21	End-of-life modules C and D are excluded due to uncertainty
D-GS-10	Liti, Audenaert & Lavagna, 2018	Journal article	Historic timber windows, glass panes, putty, seals, secondary glazing, full replacement windows	Maintenance, draught proofing, secondary glazing and replacement	Reduce life-cycle operating energy while avoiding unnecessary destructive replacement	Physical and technical obsolescence	Blower door testing, infrared thermography, dynamic simulation, technical performance decay modelling, intervention frequency testing	Assumption that replacement is always best, lack of performance decay modelling, loss of heritage value, uncertain durability of perishable components	Designers, heritage professionals, building owners, energy consultants, policymakers	Maintenance, retrofit and operational energy stage	Destructive replacement may be chosen even though less invasive measures can achieve similar operating energy savings	Implementation gap	Compare maintenance, draught proofing and secondary glazing before choosing full window replacement	Simulation- based case study	The study finds that basic maintenance and draught proofing reduce infiltration and operating energy, while internal secondary glazing can achieve energy savings similar to or greater than full replacement options	pp. 1-5, 18	The study assesses operating energy only and excludes embodied impacts
D-GS-10	Liti, Audenaert & Lavagna, 2018	Journal article	Perishable window components such as putty, sealants, paint, IGUs and timber replacement windows	Durability-aware retrofit planning	Maintain long-term energy performance by accounting for performance decay over time	Technical obsolescence	Service-life assumptions, technical performance decay, iteration frequency, modelling of initial and deteriorated states	Component deterioration, poor maintenance frequency, assuming constant performance during service life	Designers, energy modellers, building owners, maintenance planners	Use, maintenance and retrofit service life	A retrofit may appear more effective than it really is if performance decay is ignored	Lifecycle trade- off	Model both initial and deteriorated performance and plan maintenance intervals according to component service life	Dynamic simulation and technical decay modelling	The study shows that retrofit effectiveness depends on the relationship between component durability, performance decay and intervention frequency, not only on the initial energy-saving potential	pp. 2-3, 17-18	It does not provide a full environmental LCA with embodied energy or material impacts